

Reviewer Pack — Minimal Symplectic Volume and Communication Complexity Rank ...

Entry f7781080cbbe · INCONCLUSIVE

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Minimal Symplectic Volume and Communication Complexity Rank Inequality

Entry ID: f7781080cbbe **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-05 01:35:51 UTC

1. Conjecture

Field A (mathematical branch): Symplectic Geometry **Field B** (complexity object): Communication Complexity (Matrix Rank)

Statement:

For any given Boolean formula φ with n variables, the minimal symplectic volume ($SV(\varphi)$) of its associated representation in projective space is linearly correlated with its communication complexity rank ($CCrank(\varphi)$), such that $SV(\varphi) = \Omega(CCrack(\varphi))$.

Rationale (proposer's reasoning):

Symplectic geometry provides a rich algebraic structure for studying geometric properties, and its minimal symplectic volume could potentially expose underlying geometric complexities. The communication complexity rank measures the difficulty of distributing information among parties in a network, making it a suitable complexity-theoretic object. A strong correlation between these two quantities might reveal deep structural connections between geometric and communication complexity.

Taxonomy category: symplectic_complexity (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): f79cfaa3b44dca5a

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for all generated Boolean formulas φ with $n \leq 40$, the correlation coefficient between $SV(\varphi)$ and $CCrank(\varphi)$ over 30 seeds is greater than or equal to 0.5, with no seed producing a correlation coefficient less than 0.3.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction
import itertools

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_formula(n):
        clauses = []
        for _ in range(2 * n):
            clause = [random.randint(1, n), random.choice([-1, 1])]
            clauses.append(clause)
        return clauses

    def projective_volume(formula):
        n = len(formula[0])
        points = [0] * n
        for clause in formula:
            var, sign = clause
            if sign == -1:
                var = -var
            if 1 <= abs(var) <= n:
                points[var - 1] += 1
        return max(points)

    def communication_complexity_rank(formula):
        n = len(formula[0])
        rank = 0
        for clause in formula:
            var, sign = clause
            if sign == -1:
                var = -var
            if 1 <= abs(var) <= n:
                rank += 1
        return rank
```

```

n_values = [5, 10, 15, 20, 30, 40]
total_metric_value = 0
instances_tested = 0
n_max = 0
conjecture_holds = True
counterexample = ""

for n in n_values:
    for _ in range(5):
        formula = generate_formula(n)
        sv = projective_volume(formula)
        ccrank = communication_complexity_rank(formula)
        total_metric_value += sv / ccrank if ccrank != 0 else float('inf')
        instances_tested += 1
        n_max = max(n_max, n)

mean_metric_value = total_metric_value / instances_tested
correlation_coefficient = (instances_tested * sum(sv / ccrank for sv, ccrank in zip([projective_volume(f) for f in formulas], [communication_complexity_rank(f) for f in formulas])))

if correlation_coefficient < 0.3:
    conjecture_holds = False
    counterexample = "correlation_coefficient_too_low"

return {
    "metric_name": "Correlation Coefficient",
    "metric_value": correlation_coefficient,
    "instances_tested": instances_tested,
    "n_max": n_max,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {{\"seed\": {seed}, ...run_trial output...}}")
        results.append(result)

    mean_metric_value = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std=0.0 support_fraction=1.0")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std=0.0 support_fraction={support_fraction}")
    else:
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"correlation_coefficient_too_low\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_39ac7aef.py", line 87, in <module>
    result = run_trial(seed)
            ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_39ac7aef.py", line 59, in run_trial
    sv = projective_volume(formula)
        ~~~~~^~~~~~
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_39ac7aef.py", line 35, in projective_volume
    points[var - 1] += 1
    ~~~~~^~~~~~
```

IndexError: list index out of range

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test crashed before producing data, which means it did not complete its execution to provide a correlation coefficient for the conjecture. | next: Re-run the test with proper error handling and ensure that it completes without crashing.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	13988
2	preregistration	ollama_remote	glm4:latest	0	0	9437
3	novelty	ollama_remote	glm4:latest	0	0	13543
4	novelty	ollama_remote	glm4:latest	0	0	8528
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	22003
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8155
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10697
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11838
9	judge	ollama_remote	glm4:latest	0	0	12427

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 110616 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/f7781080cbbe.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/f7781080cbbe.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/f7781080cbbe.tar.gz` (if generated)